CSC341

PRACTICAL ACTIVITIES A–F

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CSC341 — Computational Science and Numerical Methods

Practical Activities A–F

The questions, MATLAB code and expected outputs are provided below in text format. Screenshots of the code and results from MATLAB are included under each activity.

PRACTICAL ACTIVITY A: Interactive Calculations

Question: Execute the following commands in the MATLAB Command Window and observe the outputs: 12 + 8, 15 − 6, 7 × 9, 45 ÷ 5, 3^4, and (12 + 8) ÷ 4.

MATLAB Code

```
12 + 8
15 - 6
7 * 9
45 / 5
3^4
(12 + 8) / 4
```

Expected Output

```
ans = 20
ans = 9
ans = 63
ans = 9
ans = 81
ans = 5
```

Questions for Reflection

1. What is the difference between = (assignment) and == (equality comparison)?

The = operator assigns a value to a variable, while == compares two values to check whether they are equal. For example, x = 5 assigns 5 to x, while x == 5 checks whether x is equal to 5.

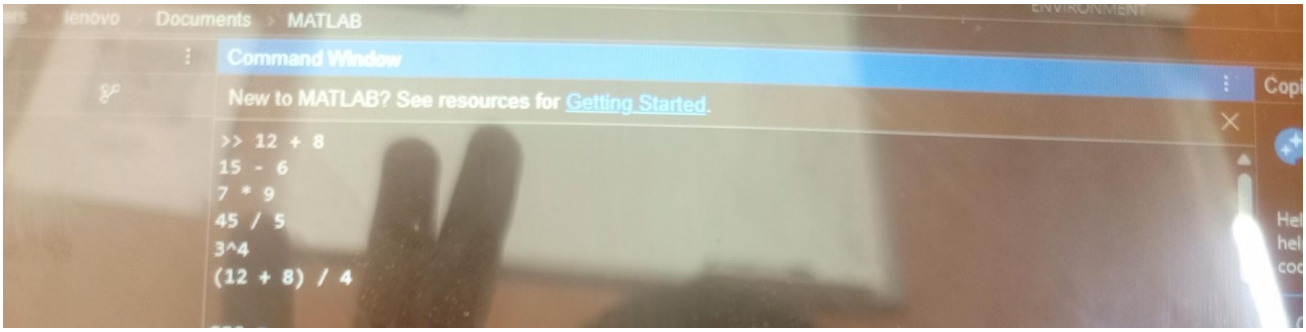
2. What happens if you remove parentheses from (12 + 8) / 4?

Without the parentheses, MATLAB follows the normal order of operations. Division is performed before addition, so 12 + 8 / 4 gives 14 instead of 5.

3. Why is suppressing output with ; essential in large numerical loops?

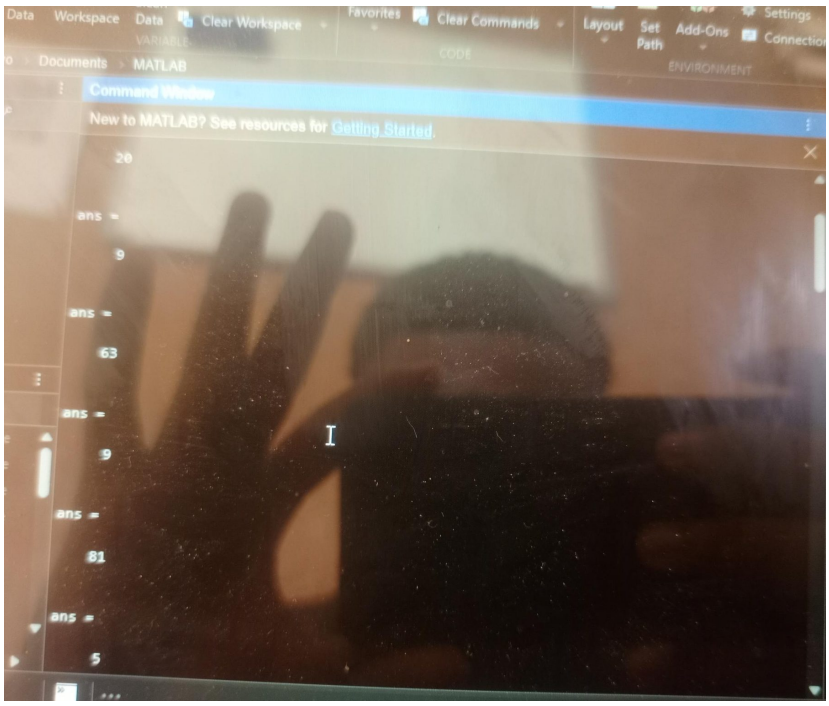
A semicolon prevents MATLAB from displaying unnecessary results in the Command Window. This reduces screen clutter and can improve performance when many calculations are performed.

Screenshot 1 — Code for Practical Activity A



MATLAB Command Window showing the commands entered for Practical Activity A.

Screenshot 2 — Output for Practical Activity A



MATLAB Command Window showing the results (ans = 20, 9, 63, 9, 81, 5).

PRACTICAL ACTIVITY B: Geometric Calculations

Question: Write a script named lab_rectangle.m that computes the perimeter P = 2(l + w) and area A = l × w for a rectangle with length l = 12 and width w = 5. Ensure your output uses formatted fprintf statements.

MATLAB Code — lab_rectangle.m

```
% lab_rectangle.m
l = 12;
w = 5;

P = 2 * (l + w);
A = l * w;

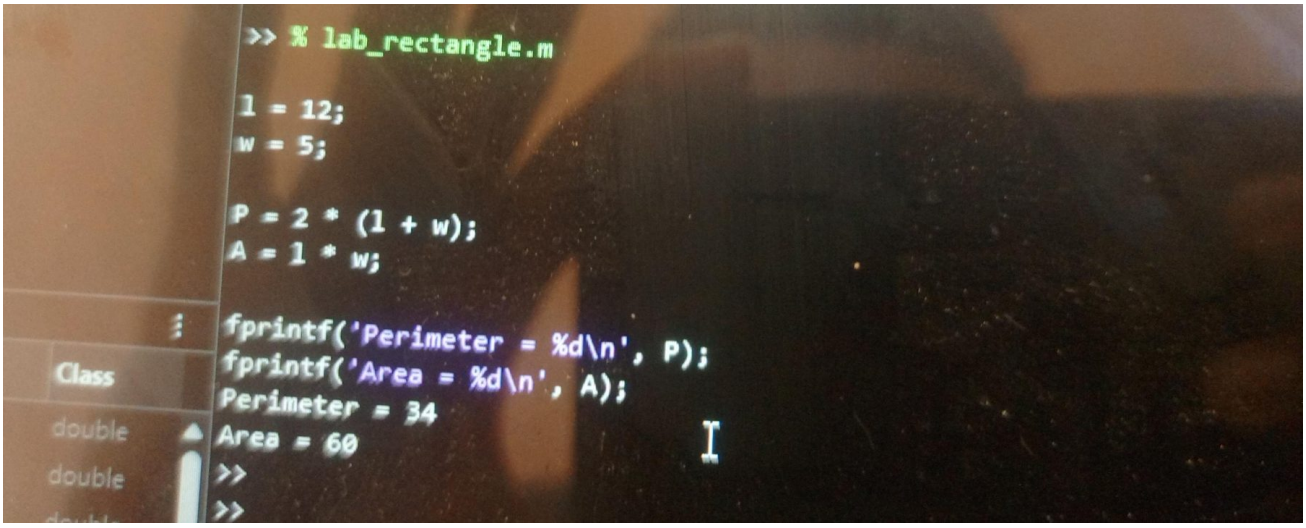
fprintf('Perimeter = %d\n', P);
fprintf('Area = %d\n', A);
```

Expected Output

```
Perimeter = 34
Area = 60
```

Calculation: Perimeter = 2 × (12 + 5) = 34 units. Area = 12 × 5 = 60 square units.

Screenshot — Code and Output for Practical Activity B (lab_rectangle.m)



MATLAB code for lab_rectangle.m with its output (Perimeter = 34, Area = 60).

PRACTICAL ACTIVITY C: Statistical Operations on Vectors

Question: Enter the vector representing student exam scores: [55, 62, 71, 84, 68, 90, 76]. Calculate the total, mean, maximum and minimum scores. Retrieve the 1st, 4th and 7th scores using vector indices.

MATLAB Code

```
scores = [55, 62, 71, 84, 68, 90, 76];
total_val = sum(scores);
mean_val = mean(scores);
max_val = max(scores);
min_val = min(scores);

fprintf('Total: %d | Mean: %.2f | Max: %d | Min: %d\n', ...
    total_val, mean_val, max_val, min_val);

first_score = scores(1);
fourth_score = scores(4);
seventh_score = scores(7);

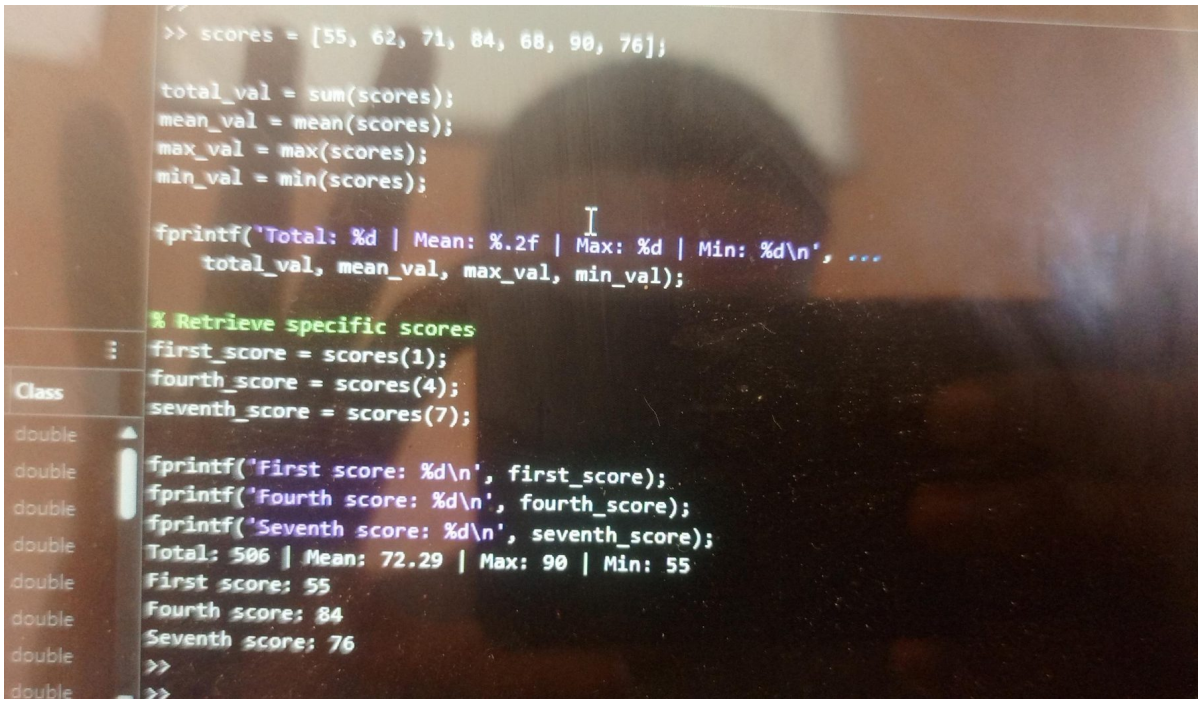
fprintf('First score: %d\n', first_score);
fprintf('Fourth score: %d\n', fourth_score);
fprintf('Seventh score: %d\n', seventh_score);
```

Expected Output

```
Total: 506 | Mean: 72.29 | Max: 90 | Min: 55
First score: 55
Fourth score: 84
Seventh score: 76
```

Explanation: sum() calculates the total, mean() calculates the average, max() finds the highest value, and min() finds the lowest value. MATLAB uses parentheses for vector indexing, such as scores(1).

Screenshot — Code and Output for Practical Activity C



MATLAB code for the exam-score statistics with its output.

PRACTICAL ACTIVITY D: Matrix Operations Comparison

Question: Execute the code and observe the differences between matrix multiplication (*) and element-wise multiplication (.*) using A = [2, 3; 4, 5] and B = [1, 6; 2, 7].

MATLAB Code

```
A = [2, 3; 4, 5];
B = [1, 6; 2, 7];

mult_standard = A * B;
mult_element = A .* B;

disp("Standard Matrix Product:");
disp(mult_standard);

disp("Element-wise Product:");
disp(mult_element);
```

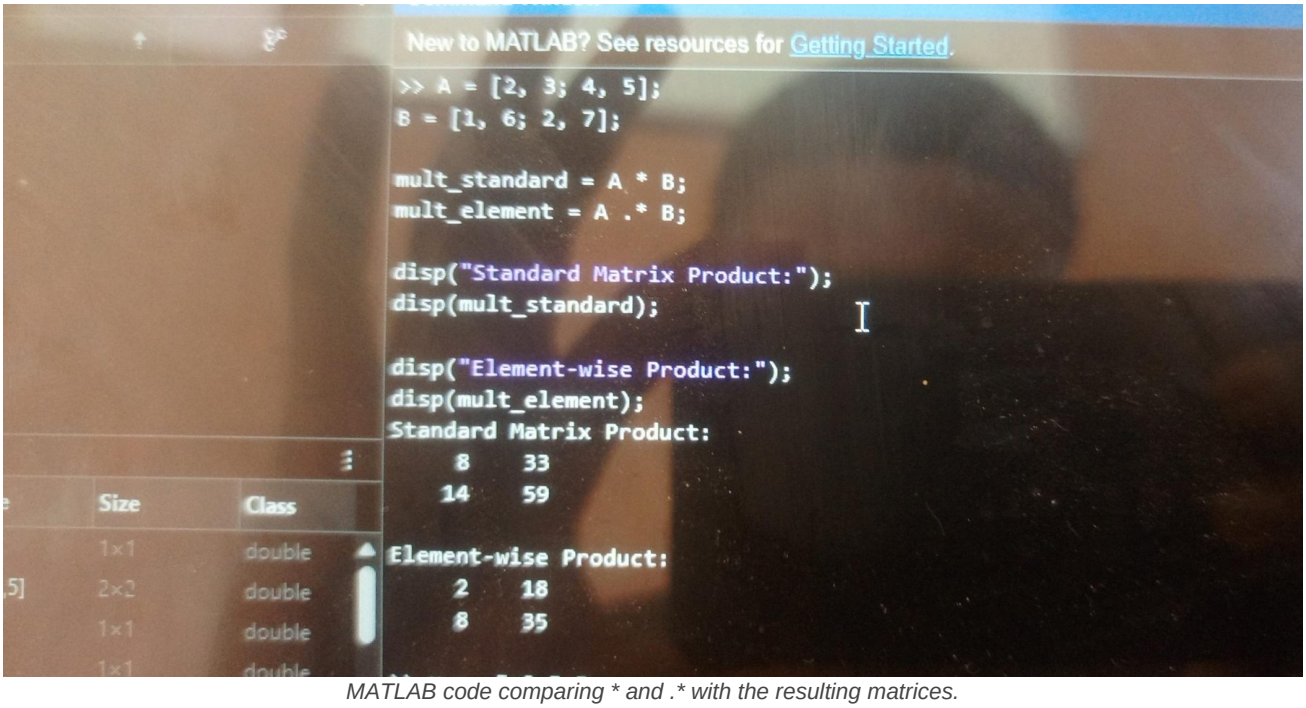
Expected Output

```
Standard Matrix Product:
     8    33
    14    59

Element-wise Product:
     2    18
     8    35
```

Explanation: The * operator performs standard matrix multiplication using rows and columns. The .* operator multiplies corresponding elements in the same positions.

Screenshot — Code and Output for Practical Activity D



*MATLAB code comparing * and .* with the resulting matrices.*

PRACTICAL ACTIVITY E: Plotting Linear Functions

Question: Create a script that plots y = 2x + 1 over −5 ≤ x ≤ 5 with a step size of 0.5. Include axis labels (xlabel, ylabel), a title, and enable grid on. Confirm visually that when x = 0, y = 1.

MATLAB Code

```
x = -5:0.5:5;
y = 2*x + 1;

figure;
plot(x, y);

xlabel('x values');
ylabel('y values');
title('Graph of y = 2x + 1');

grid on;

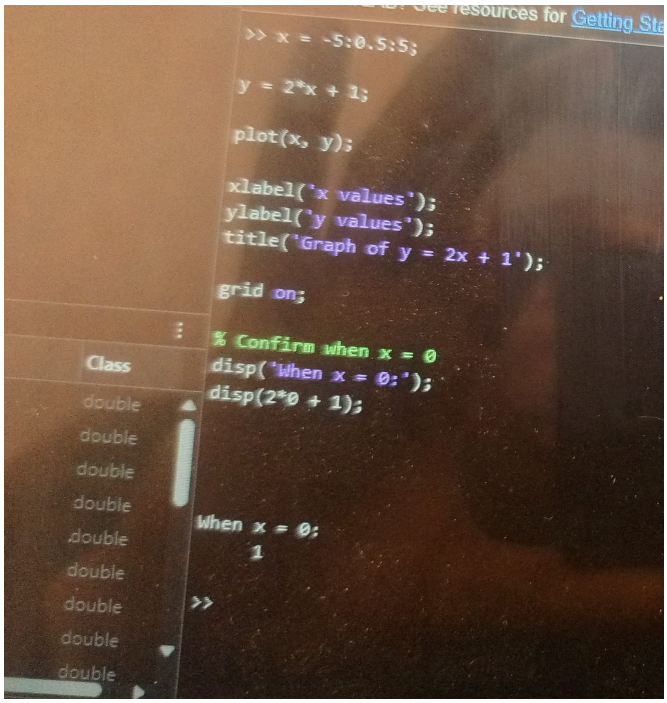
% Confirm when x = 0
disp('When x = 0:');
disp(2*0 + 1);
```

Expected Command Window Output

```
When x = 0:
     1
```

Explanation: The expression -5:0.5:5 creates x-values from −5 to 5 in steps of 0.5. The plot(x,y) command draws the graph. The graph crosses the y-axis at (0, 1), confirming that y = 1 when x = 0.

Screenshot 1 — Code and Command Window Output for Practical Activity E



MATLAB code for plotting y = 2x + 1; the Command Window prints "When x = 0:" followed by 1.

Screenshot 2 — Graph Produced by the Code (Practical Activity E)



Figure 1: Graph of y = 2x + 1, crossing the y-axis at (0, 1).

PRACTICAL ACTIVITY F: Applied Mini-Task — Lab Energy Consumption

Question: A computer laboratory houses N = 30 computers, each consuming P_computer = 150 W. The lab runs for t = 6 hours per day. Calculate the total power draw, daily energy consumption and 5-day weekly energy consumption.

Formulas:

```
P_total = N × P_computer
E_daily (kWh) = (P_total × t) / 1000
E_weekly = E_daily × days
```

MATLAB Code

```
% Lab Energy Consumption Model
N = 30;        % Number of computers
P_comp = 150;  % Watts per computer
hours = 6;     % Daily operating hours
days = 5;      % Days per week

P_total = N * P_comp;
E_daily = (P_total * hours) / 1000;
E_weekly = E_daily * days;

fprintf('Total Power Draw: %.2f W\n', P_total);
fprintf('Daily Energy Consumption: %.2f kWh\n', E_daily);
fprintf('5-Day Weekly Energy: %.2f kWh\n', E_weekly);
```

Expected Output

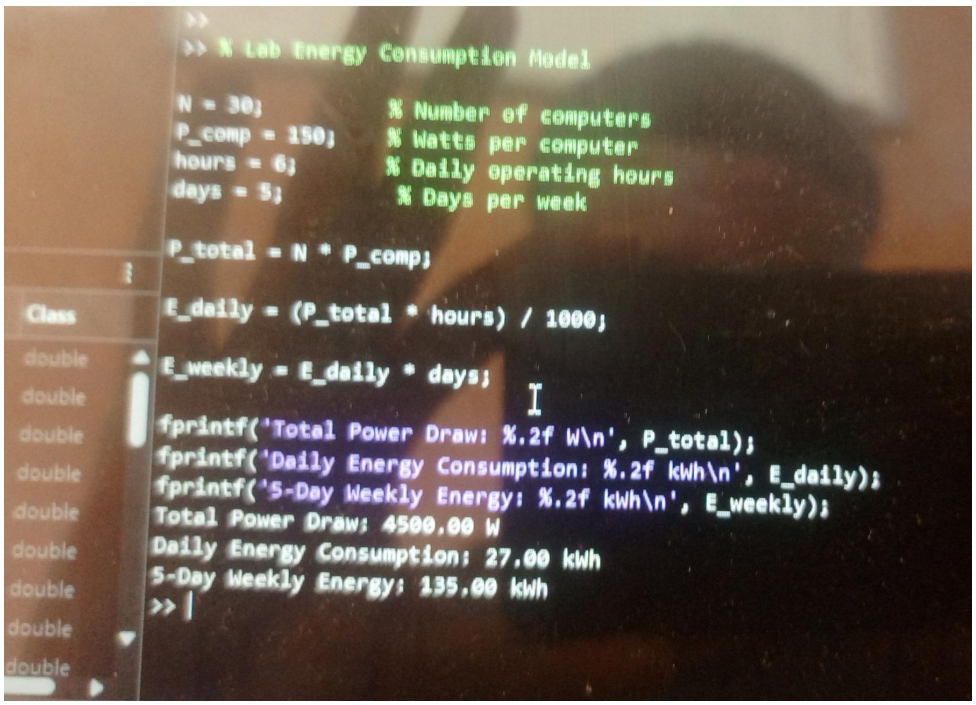
```
Total Power Draw: 4500.00 W
Daily Energy Consumption: 27.00 kWh
5-Day Weekly Energy: 135.00 kWh
```

Calculation Summary

Quantity	Calculation	Result
Total power	30 × 150	4500 W
Daily energy	(4500 × 6) / 1000	27 kWh
5-day weekly energy	27 × 5	135 kWh

Note: Save each activity as a separate .m file if your lecturer requires separate scripts. Suggested filenames: activity_A.m, lab_rectangle.m, activity_C.m, activity_D.m, activity_E.m and activity_F.m. Activity A can also be entered directly in the Command Window.

Screenshot — Code and Output for Practical Activity F



MATLAB code for the lab energy consumption model with its output (4500.00 W, 27.00 kWh, 135.00 kWh).